

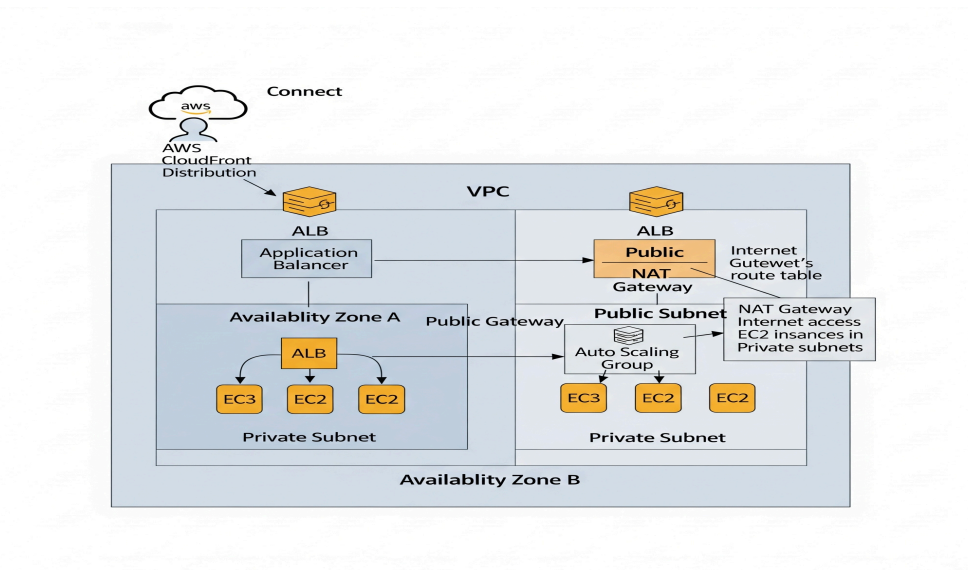
# Scalable and Low-Latency Web Architecture

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## 1. Objective

This document outlines the proposed cloud architecture for hosting a simple company website on Amazon Web Services (AWS). The design focuses on fulfilling the core requirements of a system that is **highly available**, provides **low latency** for global visitors, and features a backend that can **autoscale** based on user demand. The entire infrastructure will be managed using an Infrastructure as Code (IaC) approach to ensure it is understandable and replicable, as required by the assignment.

## 2. Architecture Diagram



## 3. Architectural Decisions

The following AWS services have been selected to meet the specific requirements of the project:

- **Amazon CloudFront**

To meet the requirement for low latency for visitors from all-around the globe, Amazon CloudFront is used. As a Content Delivery Network (CDN), it caches the website's static content at AWS edge locations worldwide. This brings the content closer to the end-users, significantly reducing load times and improving the user experience.

- **Application Load Balancer (ALB)**

To ensure the website is highly available, an Application Load Balancer is implemented. It distributes incoming traffic across multiple servers in different Availability Zones, which prevents the website from going down if a single server or an entire Availability Zone experiences an issue.

- **EC2 Auto Scaling Group**

The requirement for a backend that can autoscale is met by using an EC2 Auto Scaling Group. This service monitors the load on the EC2 instances and automatically increases or decreases the number of active web servers based on real-time visitor traffic, ensuring both performance and cost-efficiency while maintaining high availability.

- **Infrastructure as Code (IaC)**

To make the AWS infrastructure understandable and replicable, Terraform will be used as the Infrastructure as Code (IaC) tool. This allows the entire cloud environment to be defined and managed through code, which is a best practice for modern cloud deployments.